NKOSIMPHILE KHUMALO

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PROFESSIONAL SUMMARY

Final-year BSc IT student with an 80% average, currently interning as a backend developer on the Quote My Move application. My technical focus centers on microservices, low-level Linux tooling, and practical AI integrations using Go, Java and Spring Boot. Beyond the technical mechanics, I enjoy good teamwork, sharing ideas and building software that solves real problems.

TECHNICAL SKILLS

Languages: Java, Go, C++, SQL
Frameworks: Spring Boot, Gin, gRPC, Node.js, Angular
Infrastructure: Linux, Docker, AWS, GCP, GoSec, Trivy, GitHub Actions, CI/CD
Tooling: Model Context Protocol (MCP), RAG, Claude Code, Codex

EXPERIENCE

Quote My Move Dec 2025 – Present
Software Engineering Intern Johannesburg, Gauteng

- Engineered and optimized core backend API endpoints in Go, improving request handling performance and response times across key service routes.
- Integrated automated DevSecOps scanning tools (GoSec and Trivy) into GitHub Actions pipelines to identify code vulnerabilities and secure dependencies.
- Implemented frontend input validation workflows and client-side error handling to ensure seamless communication with backend services.
- Migrated environment configuration keys and API credentials to GitHub Actions Secrets, reinforcing strict pipeline and repository security.
- Collaborated directly with a senior mentor on code reviews, database migrations, and production debugging across the backend architecture.

PROJECTS

MediCare RAG System *Full-Stack AI Insurance Platform*
Stack: Java 21, Spring Boot, Node.js, React, PostgreSQL, SQLite, RAG **GitHub**

- Engineered MediCare Companion, a full-stack AI guide helping policyholders, beneficiaries and prospective members navigate complex policy terms, claims and cover estimates
- Architected a high-density RAG pipeline in Java and SQLite, indexing over 46,000 structured document chunks and vector embeddings with Gemini fallback to eliminate hallucinations.
- Implemented multi-layer AI safety controls, including pre-retrieval role and intent checks, intent-based retrieval filtering (e.g., restricting queries to death claim material), and post-generation compliance verification.
- Connected end-to-end architecture across a React 19 frontend, Node.js 22 BFF API, and Java 21 Spring Boot microservice, securing end-user sessions using HttpOnly JWT tokens and Cloudflare Turnstile.

MiniLLM Quantization Engine *Causal Transformer & Compression Engine*
Stack: Java 21, Go, Maven, INT8 Quantization **GitHub**

- Engineered a local causal Transformer training engine from scratch in Java 21, implementing auto-regressive next-token prediction and early stopping validation.

- Built a high-performance Go pipeline to tokenize raw text files, execute INT8 Post-Training Quantization (PTQ) and orchestrate Quantization-Aware Training (QAT) fine-tuning loops with the Java engine.
- Constructed an automated benchmarking suite evaluating 18,887 next-token predictions, demonstrating that INT8 QAT maintained and slightly improved top-1 prediction accuracy over FP32 (76.3% vs 72.98%).
- Designed the architecture to enable local, privacy-focused LLM training and INT8 weight compression on resource-constrained hardware without relying on cloud AI APIs.

MiniDocker Container Runtime *Lightweight Container System & Control Plane*
Stack: Go, Java 21, Spring Boot, gRPC **GitHub**

- Engineered a lightweight Linux container runtime in Go utilizing namespaces, and cgroups v2 to enforce process isolation and strict CPU/memory resource limits.
- Architected a Java Spring Boot control plane providing HTTP APIs for container lifecycle management, image archive extraction, network address allocation, and persistent state tracking
- Established low-latency Inter-Process Communication (IPC) between the Java control plane and Go runtime workers using gRPC and Protocol Buffers over local Unix domain sockets.
- Integrated an AI build assistant to recommend containerization choices for Go, Java, and Vite workloads, enforcing explicit user authorization before local tooling applies execution changes.

Linux MCP Server *Desktop AI Assistant & System Diagnostics*
Stack: Go, Wails, MCP, Linux OS **GitHub**

- Developed a Go desktop system assistant utilizing Wails to translate natural language queries into real-time Linux inspections across disk space, active ports, processes, and logs.
- Implemented Model Context Protocol (MCP) tool endpoints over STDIO, enabling external LLM clients to safely execute system diagnostic commands and inspect runtime status.
- Engineered explicit execution guardrails within the desktop GUI, requiring manual user approval for system-altering actions while blocking unauthorized commands in headless MCP mode.
- Provided developers and system administrators with a transparent, secure AI interface for monitoring and troubleshooting Linux environments without manual CLI flag execution.

EDUCATION

Richfield Graduate Institute of Technology Johannesburg, South Africa
Bachelor of Science in Information Technology | **Average:** 80% Jan 2024 – Dec 2026

- Demonstrated consistent academic excellence across core software engineering, system architecture, database administration, and cloud infrastructure modules.

National Senior Certificate (Matric) South Africa
Bachelor's Pass Endorsement Dec 2023

- **Key Subjects & Performance:** Mathematics (80%), Physical Sciences, Life Sciences, Geography

CERTIFICATIONS

AWS re/Start Graduate *Amazon Web Services, 2025*
Cloud Architecture, Linux Systems, Networking, Security & Python Automation

Build Infrastructure with Terraform on Google Cloud *Google Cloud, 2026*
Infrastructure as Code (IaC), Provisioning, Resource Management & State Modules

Build a Secure Google Cloud Network *Google Cloud, 2026*
VPC Architecture, Cloud IAM, Firewall Configuration & Perimeter Security

Create Your First Gemini Enterprise Application *Google Cloud, 2026*
Generative AI Integration, Enterprise Search, Context Engineering & LLM Workflows